

D885

Three Dimensional Science and Engineering

Armored Drill

8 concepts · 12 items · 4 transfer items

What this is. Every concept in this course's public description, re-asked with more disguise than a practice test uses. Each stem carries facts that do not decide it. Every wrong answer is a named trap.

Source. Scope from the course's public description in the WGU Institutional Catalog (September 2026); items written from the field's standard body of knowledge. No WGU assessment item, course material or proprietary content is used or reproduced. Not affiliated with or endorsed by Western Governors University.

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How to run it

Paper first. Pencil. One pass, no notes, a timer if you want the pressure.

1. **Before you read the options**, fill the strip under the item: the target word (what is actually being asked), and the one fact in the stem that decides it.
2. **Principle before arithmetic**. Name the rule, then compute.
3. **Camouflage**. Most of every stem does not matter. Ask: which single fact decides this?
4. **Inverse pairs are the bait**. If two options are opposites, one of them is the trap built for you.
5. **Mark sure, unsure or guess** honestly. Sure-and-wrong is the most valuable miss on the page — it is a wrong mental model.
6. **Absolutes** (always, never, guarantee) are usually wrong. And recognizing an answer is not the same as owning it.
7. Items with a **heavy black bar** on the left are **transfer items**: the same principle, reversed or solved for a different variable.
8. Stop at the STOP page. Then grade, fill the triage grid, and tally the traps you fell for. The biggest tally is what you drill next.

The strip, filled in, looks like this:

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	_____	

Items

1.

A new teacher plans a lesson on density. The class has 28 students and meets in a lab with six sinks. His plan: first lecture on the formula and define density, then have students float objects to confirm the rule. Which change best fits phenomena-based, 5E-style instruction?

- A. Always lecture first so students have correct words.
- B. Have students float and sink objects and look for patterns before introducing the term and formula.
- C. Keep the order but add a quiz after the lecture.
- D. Skip the hands-on part so there is more time to explain.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

2.

A student explains why a plant in a dark closet died. The student is a strong reader and likes art. She writes: the plant in the dark died, and the plant in the light grew 6 cm, so plants need light. What is missing?

- A. Reasoning that uses a science idea, such as plants using light to make sugars for energy and growth.
- B. Evidence
- C. A claim
- D. A labeled drawing, since the student likes art

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

3. [TRANSFER — reversed stem / different variable]

A student writes: my claim is that salt water freezes at a lower temperature. Reasoning: the salt water froze at -4 degrees C and the fresh water froze at 0 degrees C. What is wrong with the reasoning?

- A. It repeats the evidence instead of explaining why dissolved salt lowers the freezing point.
- B. It should include more data points instead.
- C. Nothing, because data is always enough.
- D. It has no claim.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

4.

Students design a device to keep an egg from breaking when dropped from 3 meters. The challenge is part of an after-school STEM night with pizza. The device must protect the egg, and students may spend no more than \$5 and use only paper and tape. Which is a constraint?

- A. The egg must not break
- B. The drop is from 3 meters
- C. It is part of an after-school STEM night
- D. No more than \$5 and only paper and tape

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

5. [TRANSFER — reversed stem / different variable]

In a water filter design challenge, the filter must remove at least 90% of the sediment. What is this requirement?

- A. A variable to measure
- B. A constraint
- C. A crosscutting concept
- D. A criterion

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

6.

Early in a unit on sound, students draw models showing sound as tiny particles flying from a drum to the ear. The drawings are displayed on a bulletin board near the door. Later, students see that sound does not travel through a vacuum jar. What should the teacher do with the models?

- A. Leave them as they are, since the first model is the student's answer.
- B. Have students revise their models to account for the new evidence.
- C. Replace them with the textbook diagram for students to copy.
- D. Take them off the bulletin board so the room looks neat.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

7.

A student in a class that meets in a portable building asks where the mass of a large oak tree came from, since the soil around it barely changed. The school has a large oak in its courtyard. Which crosscutting concept best fits this question?

- A. Patterns
- B. Scale, proportion, and quantity
- C. Energy and matter: flows, cycles, and conservation
- D. Cause and effect

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

8.

A teacher at a school with a new makerspace gives two tasks. Task A: find out why some bridges sway in the wind. Task B: build a model bridge that holds 5 kg and costs under \$10. The makerspace has two 3D printers. Which statement is correct?

- A. Both are science, because both use the makerspace.
- B. Both are engineering, because both involve bridges.
- C. Task A is a science question; Task B is an engineering problem.
- D. Task A is engineering; Task B is science.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

9. [TRANSFER — reversed stem / different variable]

Students are asked to design a lunch tray that keeps food warm for 20 minutes using only recycled materials. Which practice is central?

- A. Asking a scientific question
- B. Defining the problem and designing a solution
- C. Any hands-on task is an explanation task.
- D. Constructing an explanation

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

10.

Students research whether a sports drink improves memory. The class uses laptops purchased through a grant. They find a colorful company website that says it does, and a peer-reviewed study that found no effect. Which guidance fits evaluating information?

- A.** Treat both as equally reliable since both are online.
- B.** Trust the website because it looks professional.
- C.** Weigh the peer-reviewed study more heavily, because the company has a financial interest and the study reports its methods.
- D.** Reject all online sources, because none can ever be trusted.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

11.

A class in a school with a rooftop garden lists questions about plants. The garden has 20 raised beds. Which question is testable in a classroom investigation?

- A.** Does the amount of water per day affect the height of bean plants after three weeks?
- B.** Should the school grow more plants?
- C.** Why do plants exist?
- D.** Which plant is the prettiest?

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

12. [TRANSFER — reversed stem / different variable]

A student asks: do plants like music? What change makes this testable?

- A.** Keep the question; any question can be tested.
- B.** Ask which music is best.
- C.** Ask whether plants enjoy music more than silence.
- D.** Ask whether playing music for 2 hours a day changes the growth in height of bean plants over three weeks.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

STOP

Answers and dissection follow. Finish every item and every strip first.

Key and dissection

Struck-through text is the camouflage: facts that do not decide the item.

1. Answer B Concept 1

A new teacher plans a lesson on density. ~~The class has 28 students and meets in a lab with six sinks.~~ His plan: first lecture on the formula and define density, then have students float objects to confirm the rule. Which change best fits phenomena-based, 5E-style instruction?

Target word: sequence change **Decides:** Explore should come before the formal explanation.

A	Absolute language	5E sequencing puts exploring before explaining.
B	Correct	Explore first; explain after students have evidence.
C	Dropped element	Leaves the explanation first.
D	Inverted operation	Removes the exploring students need.

2. Answer A Concept 2

A student explains why a plant in a dark closet died. ~~The student is a strong reader and likes art.~~ She writes: the plant in the dark died, and the plant in the light grew 6 cm, so plants need light. What is missing?

Target word: missing CER part **Decides:** No science idea links the data to the claim.

A	Correct	The response has a claim and evidence but no scientific why.
B	Dropped element	6 cm of growth and a dead plant is evidence.
C	Dropped element	Plants need light is the claim.
D	Camouflage	Her interests do not decide what the explanation lacks.

3. Answer A Concept 2 · transfer

A student writes: my claim is that salt water freezes at a lower temperature. Reasoning: the salt water froze at -4 degrees C and the fresh water froze at 0 degrees C. What is wrong with the reasoning?

Target word: diagnose the reasoning **Decides:** It restates data.

A	Correct	Reasoning connects evidence to a science principle.
B	Adjacent category	More data adds evidence, not reasoning.
C	Absolute language	Data alone does not explain why.
D	Dropped element	The claim is clear.

4. Answer D Concept 3

Students design a device to keep an egg from breaking when dropped from 3 meters. ~~The challenge is part of an after-school STEM night with pizza.~~ The device must protect the egg, and students may spend no more than \$5 and use only paper and tape. Which is a constraint?

Target word: identify a constraint **Decides:** A limit on money or materials is a constraint.

A	Inverted operation	That is a criterion for success.
B	Confusion set	A test condition, not a limit on the design.
C	Camouflage	Event setting, not a limit on the design.
D	Correct	These limit the design.

5. Answer D Concept 3 · transfer

In a water filter design challenge, the filter must remove at least 90% of the sediment. What is this requirement?

Target word: classify the requirement **Decides:** It defines success.

A	Adjacent category	Sediment removed is measured, but the 90% target is the criterion.
B	Inverted operation	Constraints limit the design, such as cost or time.
C	Confusion set	Not a lens; it is a design requirement.
D	Correct	It states what the design must do to succeed.

6. Answer B Concept 4

Early in a unit on sound, students draw models showing sound as tiny particles flying from a drum to the ear. ~~The drawings are displayed on a bulletin board near the door.~~ Later, students see that sound does not travel through a vacuum jar. What should the teacher do with the models?

Target word: use of models **Decides:** Models are revised when new evidence arrives.

A	Inverted operation	Models are meant to change as evidence builds.
B	Correct	Revision with evidence is the heart of the modeling practice.
C	Confusion set	Copying a finished diagram skips the practice of modeling.
D	Camouflage	Display location does not decide how models are used.

7. Answer C Concept 5

A student ~~in a class that meets in a portable building~~ asks where the mass of a large oak tree came from, since the soil around it barely changed. ~~The school has a large oak in its courtyard.~~ Which crosscutting concept best fits this question?

Target word: matching CCC **Decides:** The question tracks where matter came from.

A	Adjacent category	Patterns fits repeating events, not tracking the source of mass.
B	Adjacent category	The question is not about comparing sizes.
C	Correct	Tracking the source of the tree's mass is following matter.
D	Confusion set	A common default, but the question is about where matter goes.

8. Answer C Concept 6

A teacher ~~at a school with a new makerspace~~ gives two tasks. Task A: find out why some bridges sway in the wind. Task B: build a model bridge that holds 5 kg and costs under \$10. ~~The makerspace has two 3D printers.~~ Which statement is correct?

Target word: science vs. engineering **Decides:** A explains; B designs to criteria and constraints.

A	Camouflage	The room does not decide the practice.
B	Confusion set	The topic does not decide the practice; the goal does.
C	Correct	Explaining vs. designing under criteria and constraints.
D	Inverted operation	Reverses the two practices.

9. Answer B Concept 6 • transfer

Students are asked to design a lunch tray that keeps food warm for 20 minutes using only recycled materials. Which practice is central?

Target word: practice for a design task **Decides:** A design goal with criteria and constraints.

A	Inverted operation	Engineering starts with defining a problem, not a science question.
B	Correct	It is an engineering task.
C	Absolute language	Hands-on work can be science or engineering.
D	Confusion set	Explanations answer why, not how to build.

10. Answer C Concept 7

Students research whether a sports drink improves memory. ~~The class uses laptops purchased through a grant.~~ They find a colorful company website that says it does, and a peer-reviewed study that found no effect. Which guidance fits evaluating information?

Target word: judging sources **Decides:** Consider evidence, expertise, and bias.

A	Dropped element	Drops the check for bias and evidence.
B	Confusion set	Design quality is not evidence.
C	Correct	Source credibility and bias matter.
D	Absolute language	Many online sources are reliable.

11. Answer A Concept 8

A class in a school with a rooftop garden lists questions about plants. ~~The garden has 20 raised beds.~~ Which question is testable in a classroom investigation?

Target word: testable question **Decides:** Names a changeable variable and a measurable outcome.

A	Correct	Water amount can be changed; height can be measured.
B	Confusion set	A value question, not a testable one.
C	Adjacent category	Too broad; no variable to change and measure.
D	Confusion set	Preference cannot be measured objectively.

12. Answer D Concept 8 • transfer

A student asks: do plants like music? What change makes this testable?

Target word: revise to testable **Decides:** Define variables.

A	Absolute language	Some questions need variables defined first.
B	Inverted operation	Makes it a preference question, less testable.
C	Dropped element	Still has no measurable outcome.
D	Correct	Names the variable changed and the outcome measured.

Triage grid

Write each item number in the box it belongs to.

	Right	Wrong
Sure	Owned.	Concept inversion — top priority. A wrong mental model; it survives rewording.
Unsure / guess	Recognition, not mastery. Re-drill with the transfer.	Not learned yet.

Trap tally

For every miss, find the letter you chose in the key and tally its trap. The biggest number is the pattern to drill.

Trap	What it does	In this drill	Your misses
Dropped element	Leaves out a required term or step.	6	
Wrong stem number	A figure copied from the prompt that is not the answer.	0	
Adjacent category	The right idea in the wrong bucket.	5	
Inverted operation	Direction flipped: add/subtract, cause/effect, higher/lower.	7	
Absolute language	Always, never, guarantee — where judgment applies.	5	
Confusion set	Every option is real; the test is exact role-matching.	9	
Camouflage	Answers a stem fact that was never the question.	4	

Anchor rows — the concept list

One row per concept: the rule to lock in, the trap that catches people, and the tell in the question that signals it.

#	Concept	Rule	Trap	Tell
1	Explore before explain	In the 5E model, students explore the phenomenon and gather evidence before the teacher introduces formal terms and explanations.	Starting with a lecture and then running a lab that only confirms the answer.	The order of activities in a planned lesson.
2	Claim, evidence, reasoning	A strong explanation states a claim, cites data as evidence, and uses a science idea to explain why the evidence supports the claim.	Treating restated data as reasoning, or skipping the science principle.	A student explanation missing the link between data and claim.
3	Criteria vs. constraints in engineering	Criteria are what a design must do to succeed; constraints are limits such as cost, time, or materials.	Swapping the two, or treating the budget as a criterion for success.	A design challenge that lists goals and limits.
4	Models are revised with evidence	In the modeling practice, students build initial models, use them to explain or predict, and revise them as new evidence arrives; a model is a thinking tool, not a finished picture to copy.	Treating a model as a textbook diagram students reproduce once.	A student model made early in a unit and new data that do not fit it.
5	Choosing the fitting crosscutting concept	Pick the crosscutting concept that matches the question: tracking where matter or energy goes is energy and matter; a repeating signal is patterns; a size comparison is scale, proportion, and quantity.	Defaulting to cause and effect for every question.	A student question that asks where something came from, how big, or how often.
6	Science practices vs. engineering practices	Science asks questions and constructs explanations about the natural world; engineering defines problems and designs solutions under criteria and constraints.	Calling any hands-on task engineering, or calling a design task a scientific explanation.	The goal of the task: understand why vs. build something that works.
7	Evaluating sources of information	Obtaining and evaluating information means judging a source's evidence, expertise, and bias, and checking claims against other reliable sources.	Trusting a source because it looks professional, is popular, or agrees with you.	Students comparing websites, ads, or articles about a science claim.
8	Testable questions	A testable question names a variable that can be changed and an outcome that can be measured; questions about preference or value are not testable by experiment.	Treating why or which is best questions as ready to test without defining measurable variables.	A list of student questions and a request to pick one for an investigation.

Course scope (public description): Three Dimensional Science and Engineering focuses on developing a comprehensive understanding of science and engineering pedagogical knowledge. This course is the first of three science teaching methods courses and provides a robust foundation in integrating disciplinary core ideas, crosscutting concepts, and science and engineering practices in phenomena-based curriculum and instruction.

Candidates will delve into planning learning experiences, designing instructional strategies, and utilizing phenomena-based teaching to promote engagement and understanding. Reflective practices, such as evaluating observed teaching, analyzing personal teaching methods, and reviewing course content, are emphasized to foster continuous improvement. The course will be assessed via an integrated performance assessment task, requiring candidates to demonstrate their instructional skills.